Minsung Kang

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Ph.D. Candidate, Mechanical Engineering | University of Georgia

Research Interests

Physics-Informed & Operator Learning, Uncertainty-Aware Deep Learning, Manufacturing and Industrial Data Analytics

Education

University of Georgia | Ph.D. in Mechanical Engineering *Aug. 2024 –*
Advisor: Hongyue Sun

University of Florida | M.S. in Industrial and Systems Engineering *Aug. 2021 – May 2024*
Advisor: Xiaochen Xian

Hankuk University of Foreign Studies | B.B.A. in Management Information Systems *Aug. 2020*

Columbus State University | Exchange Student *Dec. 2017*

Publications

Peer-Reviewed Publications

[J2] From Synthetic Faces to Synthetic Selves: AI-Mediated Identity Proximity in AI-Generated Human Images
Kang, M., Chang, B. H., & Lee, D.
AI & Society · 2026

[J1] EthicalFab: Toward Ethical Fabrication Process through Privacy-preserving Illegal Product Detection
Kang, M. & Sun, H.
North American Manufacturing Research Conference (NAMRC) 53 · 2025

Manuscripts Under Review

[R9] Diffusion-corrected Autoregressive Fourier Neural Operator for Droplet Evolution Prediction
Cao, J., **Kang, M.**, Sun, H., Zhou, C., Chung, J., Yue, X., & Shen, B.
IISE Transactions

[R8] Ethical and Responsible Advanced Manufacturing: Concepts, Practices, Challenges, and Future Directions
Kang, M., Lee, D., Johns, C., Gui, Z., Zhou, C., Morkos, B., Xu, W., & Sun, H.
Journal of Computing and Information Science in Engineering

[R7] Cognitive Anchoring Systems: Counteracting Social Conformity and Protecting Intellectual Independence in LLMs
Lee, D. & **Kang, M.**
Computational Intelligence

[R6] Quantum Computing Meets Human-AI Decision Conflict: A Game Theoretic Paradigm
Kang, M. & Lee, D.
Cognitive Systems Research

[R5] Beyond Visible Emotion: Hidden Emotions and the Future of Human-AI Understanding
Kang, M. & Lee, D.
Interacting with Computers

[R4] Benchmark Success, Generalization Failure: Cultural and Dataset Limits in Facial Emotion Recognition
Lee, D., **Kang, M.**, & Chang, B. H.
Image and Vision Computing

[R3] AI Psychology: The Missing Piece of the Intelligence Puzzle
Lee, D. & **Kang, M.**
Minds of Machines

[R2] AI Social Persona Drift and Peer Influence in Closed Digital Ecosystems
Lee, D. & **Kang, M.**
The Journal of Supercomputing

[R1] Analysis of Social Inequality and the Labor-Reward Paradox in an LLM-based Social Ecosystem
Lee, D. & **Kang, M.**
Information, Communication & Society

Manuscripts in Preparation

[P4] Diff-GFN: A Dual-Stream Differential Graph Fusion Network for 3D Object Classification
Kang, M., Liu, J., Wang, Y., & Sun, H.

[P3] FAPI-GRU: A Fairness-Aware Physics-Informed Recurrent Neural Network for Human Fatigue and Recovery Forecasting
Kang, M., Hajifar, S., & Sun, H.

[P2] A Framework for Real-Time, Uncertainty-Aware Detection of Critical 3D Objects
Kang, M., Xian, X., & Sun, H.

[P1] Korean-Aware Decoding for Automatic Speech Recognition (ASR): Structural and Phonological Constraints across CTC and Autoregressive Models
Lee, D. & **Kang, M.**

Conference Presentations

[T4] Calibrating Perceptual Trust in Robotic Vision: The Metacognition-based Adaptive Scanning System (MASS) — Oct. 2026
Kang, M. & Lee, D. — IACIS 2026, Pensacola, FL

[T3] Diffusion-corrected Autoregressive Fourier Neural Operator for Droplet Evolution Prediction — May 2026
Cao, J., **Kang, M.**, Sun, H., Zhou, C., Chung, J., Yue, X., & Shen, B. — IISE 2026, Arlington, TX
Best Paper Award — 2026 IISE QCRE Best Track Paper Competition

[T2] EthicalFab: Toward Ethical Fabrication Process through Privacy-preserving Illegal Product Detection — Jun. 2025
Kang, M. & Sun, H. — NAMRC 53, Clemson, SC

[T1] Brain Tumor Classification Using CNN-LSTM for MRI Images — May 2022
Kang, M. & Park, J. — IISE 2022, Seattle, WA

Grant Proposals

Awarded Allocations

Fairness Auditing of Deep Sequence Models for Wearable-Sensor Exertion Forecasting — 2026–2027
NSF ACCESS Allocations Program (*Explore Track, Project ID: MCH260145*), *200,000 Credits* — PI

Submitted (Not Funded)

Dynamic Cognitive Friction Injection (DCFI): An Adaptive Metacognitive Framework for Scalable Oversight in Multi-Agent Systems — Not Funded · 2026
Amazon Research Awards (*Agentic AI*) — Co-PI

Linguistic Rules and Personalized Voiceprints: A High-Efficiency Framework for Reliable Speech Recognition — Not Funded · 2026
Schmidt Sciences (*HAVI Program*) — Co-PI

Research Experience

MERIT: Manufacturing Empowered by Robotics, Intelligence, and Trustworthy Systems — Aug. 2024 –
Graduate Research Assistant, Athens, GA

Data Informatics of Systems Improvement and Design Laboratory (DISIDE) — Aug. 2021 – May 2024
Graduate Research Assistant, Gainesville, FL

Teaching Experience

ITEC 3235-02: Human Computer Interaction Summer 2026
ITEC 6740-01: Machine Learning
Guest Lecturer, Middle Georgia State University

ITEC 8100-02: Data Science and Analytics Spring 2026
ITEC 6730-01: Decision & Predictive Analytics
Guest Lecturer, Middle Georgia State University

Advanced Quality Management and Engineering for Business Processes Fall 2021
Teaching Assistant, University of Florida

Awards and Honors

Best Paper Award, 2026 IISE QCRE Best Track Paper Competition 2026
Diffusion-corrected Autoregressive Fourier Neural Operator for Droplet Evolution Prediction

University of Georgia Graduate Student Travel Grants 2025
Competitive grant awarded by the UGA Graduate School to fund conference travel.

Data Analytics Competition Finalist, IISE DAIS Track Data Analytics Competition 2022
Brain Tumor Classification Using CNN-LSTM for MRI Images

Korean Government Scholarship 2021 – 2024
$50,000 per academic year

Korea Student Aid Foundation (KOSAF) 2013 – 2020
Full scholarship per academic year

Mirae Asset Global Exchange Scholarship 2017
Merit-based scholarship to fund international exchange studies. $7,000

Service

2025 Center for Undergraduate Research Opportunities (CURO) Symposium Apr. 2025
Volunteer

UF INFORMS Student Chapter Apr. 2022 – Jul. 2023
President — led chapter meetings and organized a graduate-undergraduate mentoring program

Working Experience

AI Solution Center Mar. – Jun. 2020
Constructed an abnormal detection model for real-time identification of defects in assembly line processes

BostonAims (AI Platform Architecture Inc.) Jul. – Dec. 2019
scooter rebalancing, tuna price forecasting, and churn analytics

Republic of Korea Marine Corps Sep. 2014 – Jun. 2016
fulfilled South Korean national military service duty